

Assignment 6

Implementation of Error Detection using Checksum and CRC

Deadline: 17/8/26 & 19/8/26

Develop a socket programming application using C that simulates reliable data transmission using the following.

a) Checksum

Requirements

- Divide the input data into **8-bit words**
- perform **1's complement addition** with end-around carry
- generate the checksum at the sender side.
- At the receiver side, verify the received data using the checksum
- Determine whether an error has occurred.
- Display the intermediate sum, carry, checksum, and final verification result.

b) Cyclic Redundancy Check (CRC)

Requirements

- Sender must convert the given data into its binary equivalent
- Select must select a suitable generator (Ex. $G(x) = x^4 + x + 1$ i.e. 10011)
- Sender must compute the CRC checksum and append it to the data.
- Data with CRC is transmitted through TCP sockets.
- Receiver must recalculate the CRC using the same generator polynomial.
- Receiver reports whether the received frame is **error-free** or **contains errors**.
- Manually introduce bit errors before verification to demonstrate CRC error detection.